Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 1 · Exponentials and logarithms

Pure Y1 Ch 14
Exponentials and Logarithms

Chapter 14 · Original slide 2 · Starter

1) Given that $x = 3$ and $y = -1$, evaluate these expressions without a calculator.

b) 5^x c) 2^{2x-1} e) 11^{x+3y}

c) 3^y d) 7^{1-y}

5) Simplify these expressions, writing each answer as a single power.

f) $6^8 \div 6^2$ c) $\frac{2^5 \times 2^9}{2^8}$

g) $y^3 \times \left(y^9\right)^2$ d) $\sqrt{x^8}$

9) Plot the following data on a scatter graph and draw a line of best fit. Determine the gradient and intercept of the line of best fit, giving the answers to one decimal place.

x	1.2	2.1	3.5	4	5.8
y	5.8	7.4	9.4	10.3	12.8

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 3 · Chapter Overview

You have encountered exponential expressions like $y = 2^x$ before, but probably not 'the' exponential function $y = e^x$. Similarly, you will learn that the inverse of $y = 2^x$ is $y = \log_2 x$.

2: Use an interpret models that use exponential functions.

The population P of Davetown after t years is modelled using $P = Ae^{kt}$, where A, k are constants...

4: Understand the log function and use laws of logs.

Solve the equation:

$\log_2(2x) = \log_2(5x + 4) - 3$

(This is a continuation of (2))

1: Sketch exponential graphs.

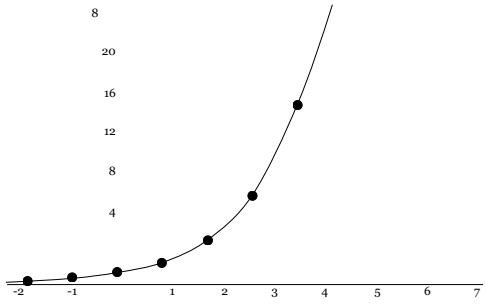
Sketch $y = 2^x$ and $y = e^x$ on the same axes.

3: Be able to differentiate e^{kx}.

If $y = 5e^{3x}$, determine $\frac{dy}{dx}$.

5: Use logarithms to estimate values of constants in non-linear models.

Chapter 14 · Original slide 4 · y={2}^{x}



x	-2	-1	0	1	2	3	4
y							

Note: Ensure that you can distinguish between a x^a (e.g. polynomial) term and an a^x exponential term. In the former the variable is in the base, and in the letter the variable is in the power. 2^x behaves very differently to x^2, both in its rate of growth (i.e. exponential terms grow much faster!) and how it differentiates.

Why are exponential functions important?

Each of the common graphs have a key property that makes them useful for modelling.

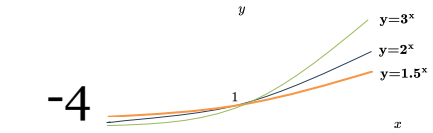
For reciprocal graphs $y = \frac{k}{x}$, as x doubles, y halves. This means we'd use it to represent variables which are inversely proportional.

Linear graphs are used when we're adding the same amount each time.

In contrast, exponential graphs are used when we're multiplying by the same amount each time. For example, we might use $1000(1.05^t)$ to model our savings with

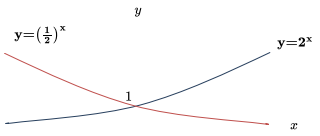
Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 5 · Contrasting exponential graphs



-4

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$



On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

Three important notes:

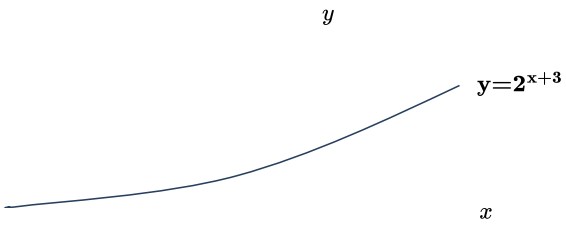
• $y = 2^x$ is said to be "exponential growing" whereas $y = \left(\frac{1}{2}\right)^x$ is said to be "exponentially decaying", because it's getting smaller (halving) each time x increases by 1.

• $y = \left(\frac{1}{2}\right)^x$ is a reflection of $y = 2^x$ in the line $x = 0$. Proof: If $f(x) = 2^x$, $f(-x) = 2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$

• $\left(\frac{1}{2}\right)^x$ would usually be written 2^{-x}.

You should therefore in general be able to recognise and sketch the graph $\mathbf{y=a^{-x}}$.

Chapter 14 · Original slide 6 · Graph Transformations



Sketch $y = 2^{x+3}$

The 'change' to 2^x is 'inside the function' (i.e. input x is replaced with $x + 3$).

Ensure you work out the new y-intercept.

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 7 · Exercise 14A (Pages 313-314)

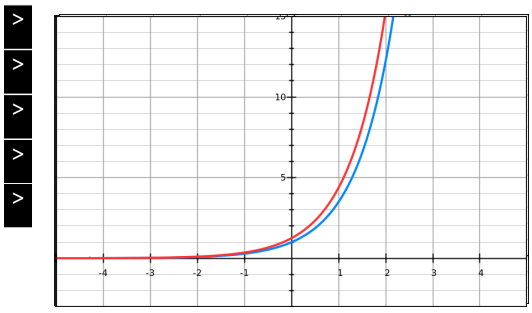
6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

Chapter 14 · Original slide 8 · Exponentials and logarithms



Click

You won't yet be able to differentiate general exponential functions $y = a^x$ until Year 2.

But I've calculated the gradient functions for you – click the black arrow to reveal the graph and gradient function.

Function

Gradient

$y = 1^x$

$\frac{dy}{dx} = 0$

>

$y = 1.5^x$

$y = 2^x$

$\frac{dy}{dx} = 0.41 \times 1.5^x$

$\frac{dy}{dx} = 0.69 \times 2^x$

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 9 · y={e}^{x}

Function	Gradient
$y = 2.5^x$ and $y = 3^x$ seem to be similar to their respective gradient functions. So is there a base between 2.5 and 3 where the function is equal to its gradient function?	$y = 1^x$
$e = 2.71828\ldots$ is known as Euler's Number.	
It is one of the five most fundamental constants in mathematics (0, 1, i, e, π).	
It has the property that:	
$\frac{dy}{dx} = 0$	>
$y = 1.5^x$	$y = 2^x$
$\frac{dy}{dx} = 0.41 \times 1.5^x$	$\frac{dy}{dx} = 0.69 \times 2^x$
>	>
$y = 2.5^x$	$\frac{dy}{dx} = 0.92 \times 2.5^x$
>	$y = 3^x$
$\frac{dy}{dx} = e^x$	$\frac{dy}{dx} = 1.10 \times 3^x$
>	>
$y = 3.5^x$	$\frac{dy}{dx} = 1.25 \times 3.5^x$
>	$y = e^x \rightarrow \frac{dy}{dx} = e^x$
Although any function of the form $y = a^x$ is known as an exponential function, e^x is known as "the" exponential function.	
You can find the exponential function on your calculator, to the right (above the "ln" key)	

Chapter 14 · Original slide 10 · Differentiating y=a{e}^{kx}

! If $y = e^{kx}$, where k is a constant, then $\frac{dy}{dx} = ke^{kx}$

Note: This is not a standalone rule but an application of something called the '*chain rule*', which you will encounter in Year 2.

$y = e^{-1x} \quad \therefore \quad \frac{dy}{dx} =$

Note: In general, when you scale the function, you scale the derivative/integral.

Different e^{5x} with respect to x.

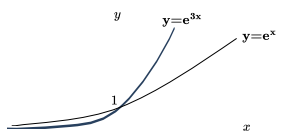
Different e^{-x} with respect to x.

Different $4e^{3x}$ with respect to x.

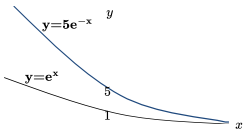
Chapter 14: Exponentials and logarithms

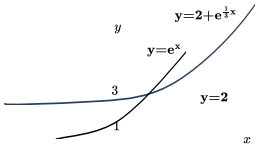
Chapter 14 · Original slide 11 · More Graph Transformations

Sketch $y = e^{3x}$



Sketch $y = 5e^{-x}$





Sketch $y = 2 + e^{\frac{1}{3}x}$

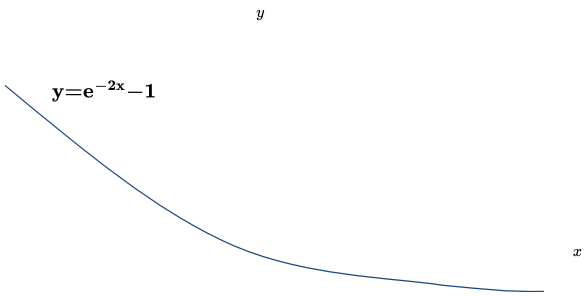
Note: Recall the shape of a $y = a^{-x}$ graph.

5 is 'outside' function, so stretch of scale factor 5 on y-axis.

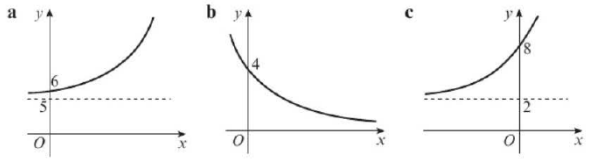
Important Note: Because the original asymptote was $y = 0$, it is now $y = 2$ and you must indicate this along with its equation.

Chapter 14 · Original slide 12 · Test Your Understanding

Sketch $y = e^{-2x} - 1$



Chapter 14 · Original slide 13 · Exercise 14B (Pages 316-317)



4 Each of the sketch graphs below is of the form $y = Ae^{bx} + C$, where A, b and C are constants. Find the values of A and C for each graph, and state whether b is positive or negative.

Chapter 14 · Original slide 14 · Exercise 14B (Pages 316-317)

6 Differentiate the following with respect to x.

a e^{6x} **b** $e^{-x/3}$ **c** $7e^{2x}$ **d** $5e^{0.4x}$ **e** $e^{3x} + 2e^{x}$ **f** $e^{x}(e^{x} + 1)$

Chapter 14 · Original slide 15 · Exercise 14B (Pages 316-317)

7 Find the gradient of the curve with equation $y = e^{3x}$ at the point where

a $x = 2$ **b** $x = 0$ **c** $x = -0.5$

8 The function f is defined as $\mathrm{f}(x) = e^{0.2x}$, $x \in \mathbb{R}$. Show that the tangent to the curve at the point $(5, e)$ goes through the origin.

Chapter 14 · Original slide 16 · Just for your interest…

Where does e come from, and why is it so important?

$e = 2.71828\ldots$

is known as Euler's Number, and is considered one of the five fundamental constants in maths: $0, 1, \pi, e, i$

Its value was originally encountered by Bernoulli who was solving the following problem:

You have £1. If you put it in a bank account with 100% interest, how much do you have a year later? If the interest is split into 2 instalments of 50% interest, how much will I have? What about 3 instalments of 33.3%? And so on…

Thus:

$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$

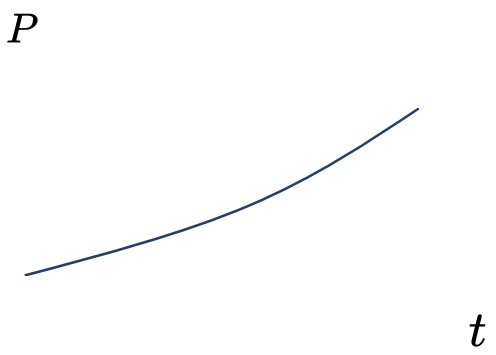
No. Instalments	Money after a year
1	$1 \times 2^1 = £2$
2	$1 \times 1.5^2 = £2.25$
3	$1 \times 1.\dot{3}^3 = £2.37$
4	$1 \times 1.25^4 = £2.44$
n	$\left(1 + \frac{1}{n}\right)^n$

But we have seen that differentiation by first principles uses 'limits'. It is therefore possible to prove from the definition above that $\frac{d}{dx}(e^x) = e^x$, and these two definitions of **e** are considered to be equivalent*.

e therefore tends to arise in problems involving limits, and also therefore crops up all the time in anything involving differentiation and integration. Let's see some applications…

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 17 · Exponential Modelling



There are two key features of exponential functions which make them suitable for population growth:

1. $\mathbf{a^x}$ gets $\mathbf{a}$ times bigger each time $\mathbf{x}$ increases by 1. (Because $\mathbf{a^{x+1}=a\times a^x}$)
With population growth, we typically have a fixed percentage increase each year. So suppose the growth was 10% a year, and we used the equivalent decimal multiplier, 1.1, as a. Then 1.1^t, where t is the number of years, would get 1.1 times bigger each year.

2. The rate of increase is proportional to the size of the population at a given moment.
This makes sense: The 10% increase of a population will be twice as large if the population itself is twice as large.

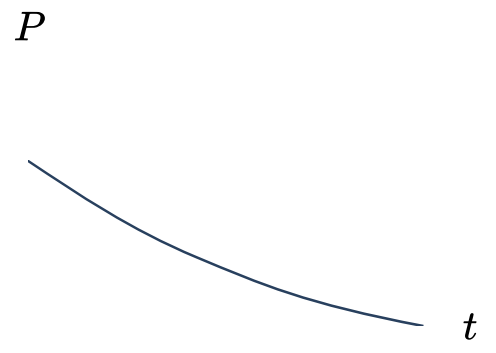
Suppose the population P in *The Republic of Nqmir* is modelled by $P = 100e^{3t}$ where t is the numbers years since *The Republic* was established.

What is the initial population?

What is the initial rate of population growth?

"Use of Technology" I use the $e^{\square}$ key. I can also use [ALPHA] [e] to get e without a power.

Chapter 14 · Original slide 18 · Another Example



[Textbook] The density of a pesticide in a given section of field, P mg/m2, can be modelled by the equation $P = 160e^{-0.006t}$

where t is the time in days since the pesticide was first applied.

a. Use this model to estimate the density of pesticide after 15 days.

b. Interpret the meaning of the value 160 in this model.

c. Show that $\frac{dP}{dt} = kP$, where k is a constant, and state the value of k.

d. Interpret the significance of the sign of your answer in part (c).

e. Sketch the graph of P against t.

a b

In general, the 'initial value', when $t = 0$, is the coefficient of the exponential term.

c d

Recall that if the power in the exponential term is negative, we have an exponential decay graph, which gradually decreases towards 0.

If in doubt, just try a large value of t.

Chapter 14 · Original slide 19 · Exercise 14C (Pages 318-319)

5 On Earth, the atmospheric pressure, p, in bars can be modelled approximately by the formula $p = e^{-0.13h}$ where h is the height above sea level in kilometres.

a Use this model to estimate the pressure at the top of Mount Rainier, which has an altitude of $4.394\,\text{km}$. (1 mark

b Demonstrate that $\frac{\mathrm{d}p}{\mathrm{d}h} = kp$ where k is a constant to be found. (2 marks

c Interpret the significance of the sign of k in part **b**. (1 mark

d This model predicts that the atmospheric pressure will change by $s\%$ for every Km gained in height. Calculate the value of s.

Chapter 14 · Original slide 20 · Exercise 14C (Pages 318-319)

6 Nigel has bought a tractor for £20 000. He wants to model the depreciation of the value of his tractor, £T, in t years. His friend suggests two models:

Model 1: $T = 20\,000e^{-0.24t}$

Model 2: $T = 19\,000e^{-0.255t} + 1000$ **a** Use both models to predict the value of the tractor after one year. Compare your results. (2 mark

b Use both models to predict the value of the tractor after ten years. Compare your results. (2 mark

c Sketch a graph of T against t for both models. (2 mark

d Interpret the meaning of the 1000 in model 2, and suggest why this might make model 2 more realistic. (1 mar

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 21 · Logarithms

You know the inverse of many mathematical operations; we can undo an addition by 2 for example by subtracting 2. But is there an inverse function for an exponential function?

Function

12

Inverse

4

$x \times 3$

4

4

7

4

$x + 3$

4

x^2

16

4

4

x^5

1024

4

4

3^x

81

4

Such functions are known as logarithms, and exist in order to provide an inverse to exponential functions.

Chapter 14 · Original slide 22 · Interchanging between exponential and log form

$$3^2 = 9 \iff \log_3 9 = 2$$

! $\log_a n$ (“said log base a of n”) is equivalent to $a^x = n$. The log function outputs the missing power.

$$\log_2 8 = 3$$

Here are two methods of interchanging between these forms.

Pick your favourite!

Method 2: Do same operation to each side of equation.

Method 1: ‘Missing Power’

• Note first the base of the log must match the base of the exponential.

• $\log_2 8$ for example asks the question “2 to what power gives 8?”

Since KS3 you’re used to the idea of doing the same thing to each side of the equation that ‘undoes’ whatever you want to get rid of.

$3x + 2 = 11$

$3x = 9$

Chapter 14 · Original slide 23 · Examples

$\log_5 25 =$

Think: "5 to the power of what gives you 25?"

$\log_3(\frac{1}{27}) =$

$\log_3 81 =$

$\log_2(\frac{1}{16}) =$

$\log_2 32 =$

$\log_{10} 1000 =$

$\log_a(a^3) =$

$\log_4 1 =$

! $\log_a 1 = 0$ for all a.

$\log_4 4 =$

$\log_4(-1) =$

$\log_2(\frac{1}{2}) =$

Chapter 14 · Original slide 24 · With Your Calculator

There are three buttons on your calculator for computing logs:

$\log_\square \square$

$\log_3 7 =$

$\ln 10 =$

$\log_5 0.3 =$

$\ln e =$

ln

ln is the "natural log of **x**", meaning "log to the base e", i.e. it the inverse of e^x.

$\ln(x) = \log_e(x)$

We will use it more extensively later this chapter.

log

$\log 100 =$

Just like the $\sqrt{}$ symbol without a number is $\sqrt[2]{\square}$ by default, log without a base is base 10 by default when used on your calculator (although confusingly "log" can mean "ln" in mathematical papers)

Chapter 14 · Original slide 25 · Exercise 14D – Y1 Page 321

6 a Without using a calculator, justify why the value of $\log_2 50$ must be between 5 and 6.

b Use a calculator to find the exact value of $\log_2 50$ to 4 significant figures.

8 a Find the values of:

i $\log_2 1$ **ii** $\log_3 1$ **iii** $\log_{17} 1$ **b** Explain why $\log_a 1$ has the same value for all positive values of a $(a \neq 1)$.

7 a Find the values of:

i $\log_2 2$ **ii** $\log_3 3$ **iii** $\log_{17} 17$ **b** Explain why $\log_a a$ has the same value for all positive values of a $(a \neq 1)$.

Chapter 14 · Original slide 26 · Extension Exercise

1

[MAT 2015 1J] Which is the largest of the following numbers?

A) $\frac{\sqrt{7}}{2}$ B) $\frac{5}{4}$ C) $\frac{\sqrt{10!}}{3(6!)}$

D) $\frac{\log_2 30}{\log_3 85}$ E) $\frac{1+\sqrt{6}}{3}$

Non-calculator!

2

[MAT 2013 1F] Three positive numbers a, b, c satisfy

$\log_b a = 2$

$\log_b(c-3) = 3$

$\log_a(c+5) = 2$

This information:

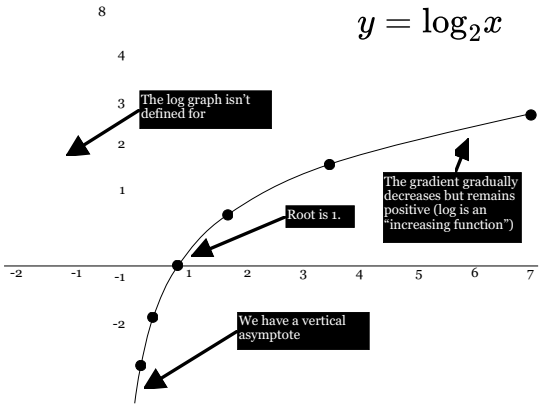
specifies a uniquely;

is satisfied by two values of a;

is satisfied by infinitely many values of a;

is contradictory

Chapter 14 · Original slide 27 · Exponentials and logarithms



x	0.25	0.5	1	2	4	8
y						

Chapter 14 · Original slide 28 · Laws of Logs

! Three main laws:

$\log_a x + \log_a y = \log_a xy$
$\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$
$\log_a \left(x^k\right) = k\log_a x$

Special cases:

$\log_a a = 1 (a > 0, a \neq 1)$
$\log_a 1 = 0 \, (a > 0, a \neq 1)$
$\log \left(\frac{1}{x}\right) = \log \left(x^{-1}\right) = -\log (x)$

Not in syllabus (but in MAT/PAT):

$\log_a b = \frac{\log_c b}{\log_c a}$

The logs must have a consistent base.

i.e. You can move the power to the front.

We often try to avoid leaving fractions inside logs.

So if the answer was:

$\log_2 \left(\frac{1}{3}\right)$

You should write your answer as: $-\log_2 3$

Reciprocating the input negates the output.

This is known as changing the base. So to get $\log_2 9$ in terms of log base 3:

$log_2 9 = \frac{log_3 9}{log_3 2} = \frac{2}{log_3 2}$

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 29 · Examples

Write as a single logarithm:

a. $\log_3 6 + \log_3 7$

b. $\log_2 15 - \log_2 3$

c. $2\log_5 3 + 3\log_5 2$

d. $\log_{10} 3 - 4\log_{10}\left(\frac{1}{2}\right)$

Write in terms of $\log_a x$, $\log_a y$ and $\log_a z$

a. $\log_a(x^2yz^3)$

b. $\log_a\left(\frac{x}{y^3}\right)$

c. $\log_a\left(\frac{x\sqrt{y}}{z}\right)$

d. $\log_a\left(\frac{x}{a^4}\right)$

Chapter 14 · Original slide 30 · Anti Laws

These are NOT LAWS OF LOGS, but are mistakes students often make:

$\log_a(b+c) = \log_a b + \log_a c$

There is no method to simplify the log of a sum, only the sum of two logs!

FAIL

$(\log_2 x)^3 = 3\log_2 x$

The power must be on the input (here the x), but here the power is around the entire log.

FAIL

Chapter 14 · Original slide 31 · Solving Equations with Logs

Solve the equation $\log_{10}4 + 2\log_{10}x = 2$

We've used the laws of logs to combine them into one.

The subtle bit: You must check each value in the original equation. We're not allowed to log a negative number.

This is a very common type of exam question.

The strategy is to combine the logs into one and isolate on one side.

Use your favourite method of rearranging. Either do "10 the power of each side" to "undo" the log, or the "insert the 2 between the 10 and $4x^2$" method.

Chapter 14 · Original slide 32 · Test Your Understanding

a

Given that $2\log_2(x+15) - \log_2 x = 6$,

(a) show that $x^2 - 34x + 225 = 0$. (5)

(b) Hence, or otherwise, solve the equation $2\log_2(x+15) - \log_2 x = 6$. (2)

b

Those who feel confident with their laws could always skip straight to this line.

Chapter 14 · Original slide 33 · Exercise 14E (323-324)

5 a Given that $\log_3(x+1) = 1 + 2\log_3(x-1)$, show that $3x^2 - 7x + 2 = 0$.

b Hence, or otherwise, solve $\log_3(x+1) = 1 + 2\log_3(x-1)$.

Chapter 14 · Original slide 34 · Exercise 14E (323-324)

Challenge:

By Writing $\log_a x = m$ and $\log_a y = n$

Prove that $\log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$

6 Given that a and b are positive constants, and that $a > b$, solve the simultaneous equations

$a + b = 13 \log_6 a + \log_6 b = 2$

Chapter 14 · Original slide 35 · Extension Exercises

These are all strictly non-calculator!

1 2

[AEA 2010 Q1b] Solve the equation

$\log_3(x-7) - \frac{1}{2}\log_3 x = 1 - \log_3 2$

[AEA 2008 Q5i] Anna, who is confused about the rules of logarithms, states that

$(\log_3 p)^2 = \log_3(p^2)$

$\log_3(p+q) = \log_3 p + \log_3 q$

However, there is a value for p and a value for q for which both statements are correct. Find their values.

3

[MAT 2007 1I] Given that a and b are positive and

$4(\log_{10} a)^2 + (\log_{10} b)^2 = 1$

what is the greatest possible value of a?

4

[MAT 2002 1F] Observe that $2^3 = 8$, $2^5 = 32$, $3^2 = 9$ and $3^3 = 27$. From these facts, we can deduce that $\log_2 3$ is:

between $1\frac{1}{3}$ and $1\frac{1}{2}$

between $1\frac{1}{2}$ and $1\frac{2}{3}$

between $1\frac{2}{3}$ and 2

none of the above

Chapter 14 · Original slide 36 · Solving equations with exponential terms

Solve $3^x = 20$

This is often said “Taking logs of both sides…”

Solve $5^{4x-1} = 61$

Chapter 14 · Original slide 37 · Solving equations with exponential terms

Solve $3^x = 2^{x+1}$

Logs in general are great for solving equations when the variable is in the power, because laws of logs allow us to move the power down.

It doesn't matter what base you use to get the final answer as a decimal, provided that it's consistent. You may as well use the calculator's 'log' (no base) key.

Why can we not apply quite the same strategy here?

This then becomes a GCSE-style 'changing the subject' type question. Just isolate x on one side and factorise out.

Chapter 14 · Original slide 38 · Test Your Understanding

1

Solve $3^{2x-1} = 5$, giving your answer to 3dp.

2

Solve $3^{x+1} = 4^{x-1}$, giving your answer to 3dp.

3

Solve $2^x 3^{x+1} = 5$, giving your answer in exact form.

Chapter 14 · Original slide 39 · Exercise 14F – Y1 Page 325

3 Solve the following equations, giving your answers to 3 significant figures where appropriate.

a $3^{x+1} = 2000$ (2 mar

b $\log_5(x-3) = -1$ (2 mar

1

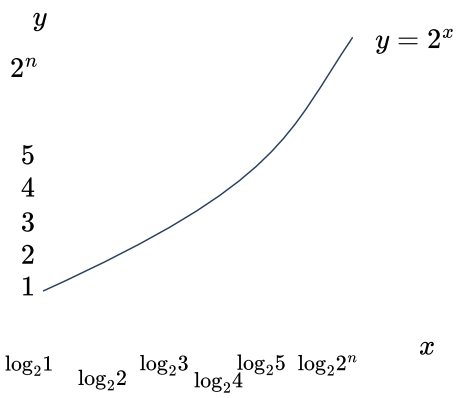
EXTENSION [MAT 2011 1H] How many positive values x which satisfy the equation:

$x = 8^{\log_2 x} - 9^{\log_3 x} - 4^{\log_2 x} + \log_{0.5} 0.25$

4 a Sketch the graph of $y = 4^x$, stating the coordinates of any points where the graph crosses the axes.

b Solve the equation $4^{2x} - 10(4^x) + 16 = 0$.

Chapter 14 · Original slide 40 · Very Challenging Extension!



2

[MAT 2013 1J] For a real number x we denote by $[x]$ the largest integer less than or equal to x. Let n be a natural number. The integral

$\int_0^n [2^x] dx$

equals:

(A) $\log_2((2^n - 1)!)$

(B) $n2^n - \log_2((2^n)!)$

(C) $n2^n$

(D) $\log_2((2^n)!)$

Chapter 14 · Original slide 41 · Natural Logarithms

We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.

We also saw that e^x is "the" exponential function.

The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

! The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x =$

$e^{\ln x} =$

Since "e to the power of" and "ln of" are inverse functions, they cancel each other out!

Solve $e^x = 5$

"ln both sides".

On the LHS it cancels out the "e to the power of"

Solve $2 \ln x + 1 = 5$

Do "e to the power of" each side. On the LHS it cancels out the ln.

Chapter 14 · Original slide 42 · Quadratics in {e}^{x}

In previous chapters we've already dealt with quadratics in disguise, e.g. "quadratic in sin". We therefore just apply our usual approach: either make a suitable substitution so the equation is then quadratic, or (strongly recommended!) go straight for the factorisation.

Solve $e^{2x} + 2e^x - 15 = 0$

Solve $e^x - 2e^{-x} = 1$

First write negative powers as fractions:

Note that $e^{2x} = (e^x)^2$ therefore

Chapter 14 · Original slide 43 · Test Your Understanding

Solve $\ln(3x+1) = 2$

Solve $e^{2x} + 5e^{x} = 6$

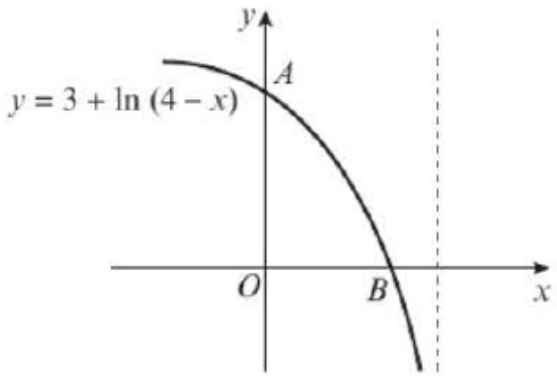
Solve $2^{x}e^{x+1} = 3$ giving your answer as an exact value.

Chapter 14 · Original slide 44 · Exercise 14G – Y1 Page 327

4 Find the exact solutions to the equation $e^{x} + 12e^{-x} = 7$.

6 Solve $3^{x}e^{4x-1} = 5$, giving your answer in the form $\dfrac{a + \ln b}{c + \ln d}$

Chapter 14 · Original slide 45 · Exercise 14G – Y1 Page 328



8 The graph of $y = 3 + \ln(4 - x)$ is shown to the right.

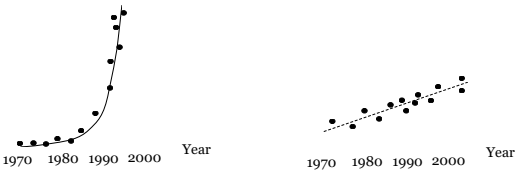
a State the exact coordinates of point A. (1

b Calculate the exact coordinates of point B. (3 m

Challenge: The graph of the function

passes through (0,5) and (6,10). Given that the line $y = 2$ is an asymptote to the graph, show that $B = \frac{1}{6}\ln\left(\frac{8}{3}\right)$

Chapter 14 · Original slide 46 · Graphs for Exponential Data



In Science and Economics, experimental data often has exponential growth, e.g. bacteria in a sample, rabbit populations, energy produced by earthquakes, my Twitter followers over time, etc.

Because exponential functions increase rapidly, it tends to look a bit rubbish if we tried to draw a suitable graph:

But suppose we took the log of the number of transistors for each computer. Suppose the number of transistors one year was y, then doubled 2 years later to get $2y$.

When we log (base 2) these:

$y \to \log_2 y$
$2y \to \log_2(2y) = \log_2 2 + \log_2 y$
$= 1 + \log_2 y$

The logged value only increased by 1! Thus taking the log of the values turns exponential growth into linear growth (because each time we would have doubled, we're now just adding 1), and the resulting graph is a straight line.

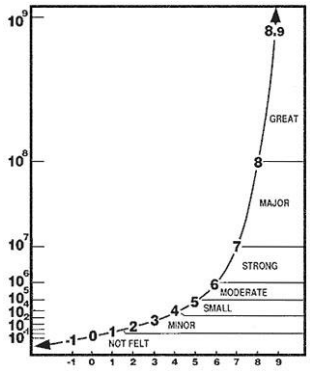
Take for example "Moore's Law", which hypothesised that the processing power of computers would double every 2 years. Suppose we tried to plot this for computers we sampled over time:

If we tried to force all the data onto the graph, we would end up making most of the data close to the horizontal axis. This is not ideal.

log(transistors)

Number of transistors

Chapter 14 · Original slide 47 · Graphs for Exponential Data



Richter Scale

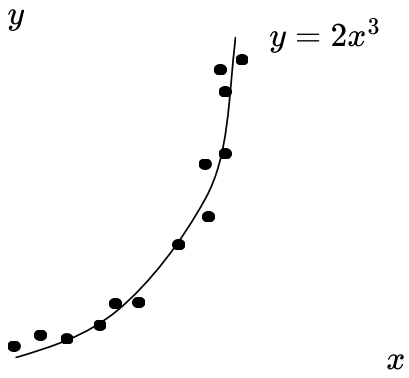
Because the energy involved in earthquakes decreases exponentially from the epicentre of the earthquake, such energy values recorded from different earthquakes would vary wildly.

The Richter Scale is a logarithmic scale, and takes the log (base 10) of the amplitude of the waves, giving a more even spread of values in a more sensible range.

(The largest recorded value on the Richter Scale is 9.5 in Chile in 1960, and 15 would destroy the Earth completely – evil scientists take note)

The result is that an earthquake just 1 greater on the Richter scale would in fact be 10 times as powerful.

Chapter 14 · Original slide 48 · Other Non-Linear Growth



We would also have similar graphing problems if we tried to plot data that followed some polynomial function such as a quadratic or cubic.

We will therefore look at the process to convert a polynomial graph into a linear one, as well as a exponential graph into a linear one...

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 49 · Turning non-linear graphs into linear ones

Case 1: Polynomial → Linear

Suppose our original model was a polynomial one*:

$y = ax^n$

Then taking logs of both sides:

$\log y = \log ax^n$
$\log y = \log a + n \log x$

We can compare this against a straight line:

$Y = mX + c$

! If $y = ax^n$, then the graph of $\log y$ against $\log x$ will be a straight line wih gradient n and vertical intercept $\log a$.

$\log y$

$\log a$

$\log x$

* We could also allow non-integer n; the term would then not strictly be polynomial, but we'd still say the function had "polynomial growth".

In summary, logging the y-axis turns an exponential graph into a linear one. Logging both the x and y-axis turns a polynomial graph into a linear one.

Case 2: Exponential → Linear

Suppose our original model was an exponential one:

$y = ab^x$

Then taking logs of both sides:

$\log y = \log ab^x$
$\log y = \log a + x \log b$

Again we can compare this against a straight line:

$Y = mX + c$

! If $y = ab^x$, then the graph of $\log y$ against x will be a straight line with gradient $\log b$ and vertical intercept $\log a$.

$\log y$

$\log a$

x

The key difference compared to Case 1 is that we're only logging the **y** values (e.g. number of transistors), not the x values (e.g. years elapsed). Note that you do not need to memorise the contents of these boxes and we will work out from scratch each time...

Chapter 14 · Original slide 50 · Example

[Textbook] The graph represents the growth of a population of bacteria, P, over t hours. The graph has a gradient of 0.6 and meets the vertical axis at $(0, 2)$ as shown.

A scientist suggest that this growth can be modelled by the equation $P = ab^t$, where a and b are constants to be found.

a. Write down an equation for the line.

b. Using your answer to part (a) or otherwise, find the values of a and b, giving them to 3 sf where necessary.

c. Interpret the meaning of the constant a in this model.

2

Equation of straight line is $y = mx + c$ where here:

Recall that $\log a$ means $\log_{10} a$

$\log(P)$

t

a b c

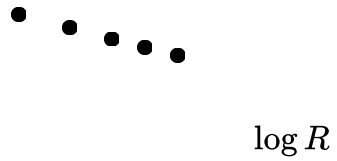
Just like on previous slide, start with the model then log it:

Comparing with our straight line equation in (a):

Recall that the coefficient of an exponential term gives the 'initial value'.

Chapter 14: Exponentials and logarithms

Chapter 14 · Original slide 51 · Example



Textbook Error: They use $R = aP^n$ but then plot $\log P$ against $\log R$.

logR				
logP				
6.4				
6.0				
5.6				
5.2				

[Textbook] The table below gives the rank (by size) and population of the UK's largest cities and districts (London is number 1 but has been excluded as an outlier).

City B'ham Leeds Glasgow Sheffield Bradford

Rank, **R** 2 3 4 5 6

Population, **P** 1 000 000 730 000 620 000 530 000 480 000

The relationship between the rank and population can be modelled by the formula:

$P = aR^n$ where a and n are constants.

a) Draw a table giving values of $\log R$ and $\log P$ to 2dp.

b) Plot a graph of $\log R$ against $\log P$ using the values from your table and draw the line of best fit.

c) Use your graph to estimate the values of a and n to two significant figures.

c

First get equation of straight line:

As with previous example, let's log the original model so we can compare against our straight line:

Comparing this with our straight line:

a b

$\log(P)$

0.2 0.4 0.6 0.8

Chapter 14 · Original slide 52 · Test Your Understanding

Notes

1. We want to find the parameters of a model, e.g. $P = ab^t$ that best fits the data (in this case the parameters we want to find are a and b).

2. If the data had a linear trend, then this would be easy! We know from KS3 that we'd just plot the data, find the line of best fit, then use the gradient and y-intercept to work out the m and c in our linear model.

3. But the original data wasn't linear, and it would be harder to draw an 'exponential curve of best fit'.

4. We therefore log the model so that the plotted data then roughly forms a straight line, and then we can then draw a (straight) line of best fit.

5. The gradient and y-intercept of this line then allows us to estimate the parameters a and b in the original model that best fit the data.

The process of finding parameters in a model, that best fits the data, is known as regression.

t	0.7	1.3	2.2
logP			

A website wants to predict his number of Twitter followers P, t years from the start 2015. He predicts that his followers will increase exponentially according to the model $P = ab^t$, where a, b are constants that he wishes to find.

He records his followers at certain times. Here is the data:

Years **t** after 2015: 0.7 1.3 2.2

Followers **P**: 2353 3673 7162

a) Draw a table giving values of t and $\log P$ (to 3dp).

b) A line of best fit is drawn for the data in your new table, and it happens to go through the first data point above (where $t = 0.7$) and last (where $t = 2.2$).
Determine the equation of this line of best fit. (The y-intercept is 3.147)

c) Hence, determine the values of a and b in the model.

d) Estimate how many followers the website will have at the start of 2020 (when $t = 5$).

Chapter 14 · Original slide 53 · Exercise 14H Page 331-333

5 Kleiber's law is an empirical law in biology which connects the mass of an animal, m, to its resting metabolic rate, R. The law follows the form $R = am^b$, where a and b are constants.

The table below contains data on five animals.

Animal	Mouse	Guinea pig	Rabbit	Goat	Cow
Mass, m (kg)	0.030	0.408	4.19	34.6	650
Metabolic rate R (kcal per day)	4.2	32.3	195	760	7637

a Copy and complete this table giving values of $\log R$ and $\log m$ to 2 decimal places. (1 m

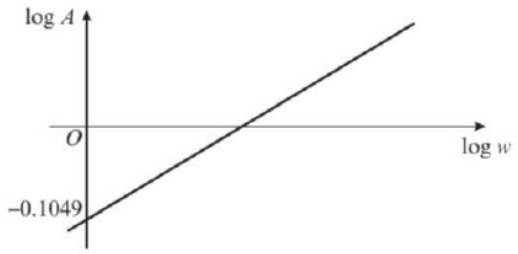
$\log m$	-1.52				
$\log R$	0.62	1.51	2.29	2.88	3.88

b Plot a graph of $\log R$ against $\log m$ using the values from your table and draw in a line of best fit.

c Use your graph to estimate the values of a and b to two significant figures.

d Using your values of a and b, estimate the resting metabolic rate of a human male with a mass of $80\,\text{kg}$.

Chapter 14 · Original slide 54 · Exercise 14H Page 331-333



From looking at this graph, the scientist suggests that the number of sick people can be modelled by the equation $N = ab^t$, where a and b are constants to be found.

The graph passes through the points $(0, 1.6)$ and $(10, 2.55)$.

a Write down the equation of the line. (2

b Using your answer to part **a** or otherwise, find the values of a and b, giving them to 2 significant figures. (4

c Interpret the meaning of the constant a in this model. (

d Use your model to predict the number of sick people after 30 days.

8 A scientist is modelling the number of people, N, who have fallen sick with a virus after t days.